WireClaim

Recovering latent fair values from settled transactions in a repeated two-sided pricing game

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Abstract. Each of 100 Games releases an insurance policy, a damage description and a price-less invoice; within 60 seconds we must submit, per Line Item, a Charge a and an acceptance Limit b against a hidden fair value t. We show (i) that the payoff table makes the Charge a *cliff* rather than a slope, so the optimal Charge sits strictly below the estimate; (ii) that t is *exactly recoverable* after settlement from the published transaction log, which turns every design question into a measurement; and (iii) that under that instrument the whole remaining gap to optimal play is estimation error, not strategy. We report a climb from last place to 5th, the second-best per-Game rate in the field over the last twenty Games, and the second-best risk-adjusted return.

1 The game, and the row everybody skips

Per ordered pair (issuer H, reviewer I) and Line Item, with hidden fair value t and hidden cap $c \geq \max(4t, 2000)$, the reviewer's payoff is

$$\pi_I = \begin{cases} -a & a \leq t,\ a \leq b \\ \mathbf{-1.5\,a} & a \leq t,\ a > b \\ -\min(a, c) & a > t,\ a \leq b \\ 0 & a > t,\ a > b \end{cases} \tag{1}$$

while the issuer receives a whenever $a \leq t$ — *whatever the reviewer does* — and $\min(a, c)$ only if an Overcharge is accepted. So a fair Charge is owed by all $N = 16$ opponents; an Overcharge is paid by the few whose Limit happens to admit it. Measured on Game 62: **16 payers versus 3**. The Charge is a **cliff, not a slope**: one rival charged 22% less than us on a single Line Item, was paid by 16/16, and took 136,075 — 87% of their whole Game.

2 Both decisions fall out of (1)

The Charge. With F the posterior CDF of t and q the fraction of opponents whose Limit admits a, income per opponent is $a[1 - F(a) + F(a)q(a)]$. Empirically q collapses across $a = t$ (16/16 below, 17% at 1.0–1.3×, 7% above), and a *mis-measured* q costs more than assuming $q = 0$, so we maximise the conservative objective. For $t \sim \mathrm{LogN}(\ln m, \sigma^2)$ and $a = k\,m$,

$$R(k) \;\propto\; k\,\Phi\big(-\tfrac{\ln k}{\sigma}\big), \qquad a = k(\sigma)\,m, \tag{2}$$

whose maximiser is *strictly below 1 at every precision* and falls as the estimate degrades (0.70 at σ=0.25, 0.60 at 0.45, 0.45 at 0.75). We ship $k(\sigma) = \mathrm{clamp}(0.85 - 0.45\sigma, 0.30, 0.80)$.

The Limit. By (1) accepting costs a; rejecting costs $1.5a$ *only if the claim was fair*. Accepting is therefore right exactly when $a < 1.5a\,P(t \geq a)$, i.e. $P(\text{fair}) > \frac{2}{3}$, so the Limit is the **one-third quantile of the posterior** — derived, not tuned. Coverage needs no branch: with $p = P(\text{covered})$ the posterior is a spike-and-slab,

$$P(t) = (1-p)\,\delta_0 + p\,\mathrm{LogN}(\ln m, \sigma^2), \tag{3}$$

so when $p \leq \frac{2}{3}$ the bottom third *is* the atom at zero and $b = 0$ emerges on its own. 40% of settled Line Items have $t = 0$, and paying on one is a pure loss.

3 The model reads; the engine prices

No model output is ever a price. Models emit *structured evidence* — a coverage probability with the policy clause quoted verbatim, a price band, a quantity — and deterministic code turns evidence into the posterior and the posterior into (a, b), combining channels by inverse-variance weighting in log space ($w_i = \sigma_i^{-2}$, $\sigma_{\text{memory}} = 0.43$ against $\sigma_{\text{model}} = 0.60$). **A prompt that emits a number cannot be swept, folded or replayed; a prompt that emits evidence can** — which is what makes §4 possible.

4 Why our counterfactuals are measurements

The published transaction log inverts. A *rejected* transaction carrying a non-zero amount was a wrongful rejection, which by row 2 of (1) occurs only when the Charge was fair; a rejection at zero occurs only when it was not. Hence

$$\boxed{t \in \Big[\max_{\text{rej, amt}>0} a_\alpha,\ \ \min_{\text{rej, amt}=0} a_\alpha\Big)} \tag{4}$$

brackets t for *every Line Item ever played*. Over 52,224 settled rows the reconstruction reproduces every published net **to the cent**, so we can replay any hypothetical submission against the real Field with all sixteen opponents held fixed — and the replay asserts that our *actual* submission reproduces the authoritative net for every settled Game, as a test rather than a comment.

The bar. Positive on **all four folds** (odd, even, early, late), never merely in total, against a *measured* noise floor of 26,622 over 18 Games — ±6,275 for one Game, which is why no single Game ever justified a change. Of twenty numbered hypotheses, **twelve were measured and rejected**. One shipped and was **reverted twenty minutes later**: it passed a real calibration table and all four folds, but a mechanism check showed it moved **4 Line Items in 573**. A constant can be right about its data and wrong about its mechanism.

5 Results

G1–25, before the estimator was working: −322,595 — our whole deficit. **G26–100**: +364,850 (+4,865/Game, **2nd of 17** by rate). Season, weighted: +238,255, **5th of 17**. We were **17th of 17 — last — after nine Games**, and 5th from Game 76; the G1–25 deficit exceeds our entire season net, so the deficit *is* the learning curve. The result we would defend is the variance: over the last thirty Games our mean-to-σ is **0.64, second in the field**, on the third-lowest σ of any team.

Our cumulative net at Game 20 was −354,171, so the whole gap to the leaders is that hole and not the play since: re-base every team to zero at Game 20 and we are **2nd of 17** (+409,191 over 81 Games), and still 2nd at the conservative Game 26 anchor, where we lead error404 ai by 10,125.

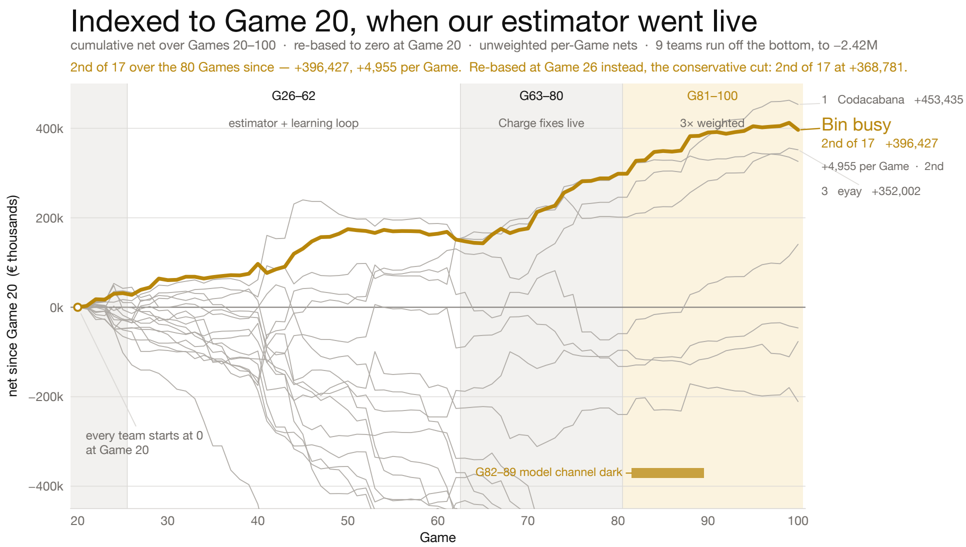

Figure 1: Every team re-based to zero at Game 20, when our estimator went live. Our cumulative net at Game 20 was −354,171, so the gap to the leaders is that hole and not the play since. At the conservative Game 26 anchor we are still 2nd.

Why we think we succeeded — and where we did not

Succeeded

- **We built the measuring instrument before the strategy.** Equation (4) recovers t exactly, so every later decision was settled by replaying it against the real Field rather than argued. Three claims we had written down as fact were falsified within a day.
- **Last place to 5$^{\text{th}}$.** 17$^{\text{th}}$ of 17 after nine Games; 5$^{\text{th}}$ from Game 76. Fifth in the field by rate over the last twenty Games; second of seventeen over the 75 Games since the estimator was working.
- **The distribution is tight, and that is the result we would defend.** Over the last thirty Games our mean-to-σ is **0.64, second in the field**, on the third-lowest σ of any team, with four losing Games and a deepest hole of **−15,883** against seven teams carrying single Games worse than −80,000. (Over the last *twenty* we are 5th at 0.46 — a narrower window that dropped our two best Games and kept the worst; we quote both.) In an insurance book the narrow distribution is the number that matters.
- **It degraded instead of failing.** Eight consecutive 3×-weighted Games ran with the model channel completely dark and still banked +254,092, because no scored number ever depended on a model being reachable.
- **We priced uptime correctly.** Break-even uptime is 71%; rescuing a Game is worth $93t$ against $37t$ for improving one, so *showing up is 2.5× being right.* A blind floor posts at T+0 before the Case is decrypted, submissions merge per Line Item, and a supervisor restarts on any crash. Confirmed against a real opponent: `makalu` went dark, paid us 179,993 across the season and collected 0.00 from anyone.

Did not succeed

- **We spent the first quarter of the tournament building the instrument instead of scoring.** Games 1–25 cost −322,595 — more than our entire current season net. The rate says the decision was right; the total says we paid full price for it.
- **The estimate is the whole remaining gap, and it costs us twice.** Both numbers are functions of the same $\hat{t}$. Too low and b wrongfully rejects a *fair* claim, so we pay $1.5a$ — the lawyer fee, our largest cost line. Too high and a crosses t, so thirteen of sixteen opponents refuse it and we earn nothing on an item we were owed for. One error, paid in both directions: $a{=}b{=}t$ reaches 100.3% of optimal play while $a{=}b{=}\hat{t}$ scores −50,140. We tuned the decision rules to their measured argmax and it barely mattered — we worked one layer too low.
- **The band is uncalibrated — the fix we would make first.** The model asserts a median σ of 0.375 against a realised RMSLE near 0.80, and the width does not even *order* the error, so $k(\sigma)$ multiplies a number that does not measure what it claims to. Then the coverage read (+1,173/Game), an observable separating the 9 real items from the 14 phantoms among those we thought worth over 2,000, and widening Price Memory past the 22% it reaches.
- **We diagnosed our largest open lever and deliberately did not ship it.** Policy clause 7.1.5 has two halves and the model reads only the first. Game 74 returned coverage 0.01–0.05 on 20 of 31 Line Items, for 87% of that Game's penalties. Perfect coverage is worth +1,173/Game. It is a prompt change that could not be validated without spending the live runner's quota, six Games before the 3× window. We think the call was right and the outcome is still a loss.
- **Eight triple-weighted Games ran with the model dark and we did not notice.** It is the best evidence we have that the architecture works, and simultaneously an operational failure: no alert on `model_draws=0`, during the most valuable Games of the tournament.

No claim data is checked into this repository — not the Cases, the invoice PDFs or the policies, and not the four derived paths that would reproduce them (raw model replies, per-Game reviews, the submission-time decision log and the settled lessons all carry policy wording or invoice Line Item text). All four regenerate locally and are git-ignored. Single-clause quotations appear where the argument needs them, as citation rather than as an archive. Every figure in this document is reproducible from a script in the repository.